



MedFlow



Project Report

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Project Introduction

MedFlow is a healthcare appointment-management platform designed to bridge the gap between patients and doctors through a seamless, digital-first experience. Built as a responsive Angular web application backed by a Spring Boot REST API and PostgreSQL database, MedFlow empowers patients to book, manage, and track their medical appointments online — eliminating the friction of phone queues, repeated paperwork, and missed reminders.

For healthcare providers, MedFlow delivers a unified doctor dashboard that consolidates schedules, patient profiles, and appointment history into a single, intuitive interface — reducing administrative overhead and preventing double-bookings.

The platform supports two distinct user roles — Patient and Doctor — each with a tailored portal experience. Key capabilities include: Google OAuth authentication, a calendar-based slot picker, real-time notifications, doctor discovery with specialty filtering, patient profile management, and fully responsive layouts that adapt seamlessly between mobile and desktop viewports.

MedFlow was developed as the Mobile Computing course project for Spring 2026 at the Faculty of Engineering, Helwan University (FEHU), following a rigorous UX-driven process from research and empathy mapping through wireframing, design system definition, and full-stack implementation.

UX Phase

The UX phase covers research, problem definition, solution ideation, and wireframing. MedFlow is a healthcare appointment-management platform connecting patients with doctors through a clean, mobile-first Angular web app backed by a REST API.

Stage I — Research

Empathy Map

Two personas were mapped: Ahmed Hassan (Patient, 32) and Dr. Karim Nour (Doctor, 45). The empathy map surfaces what each user thinks, feels, sees, hears, says and does, plus their core pains and gains.

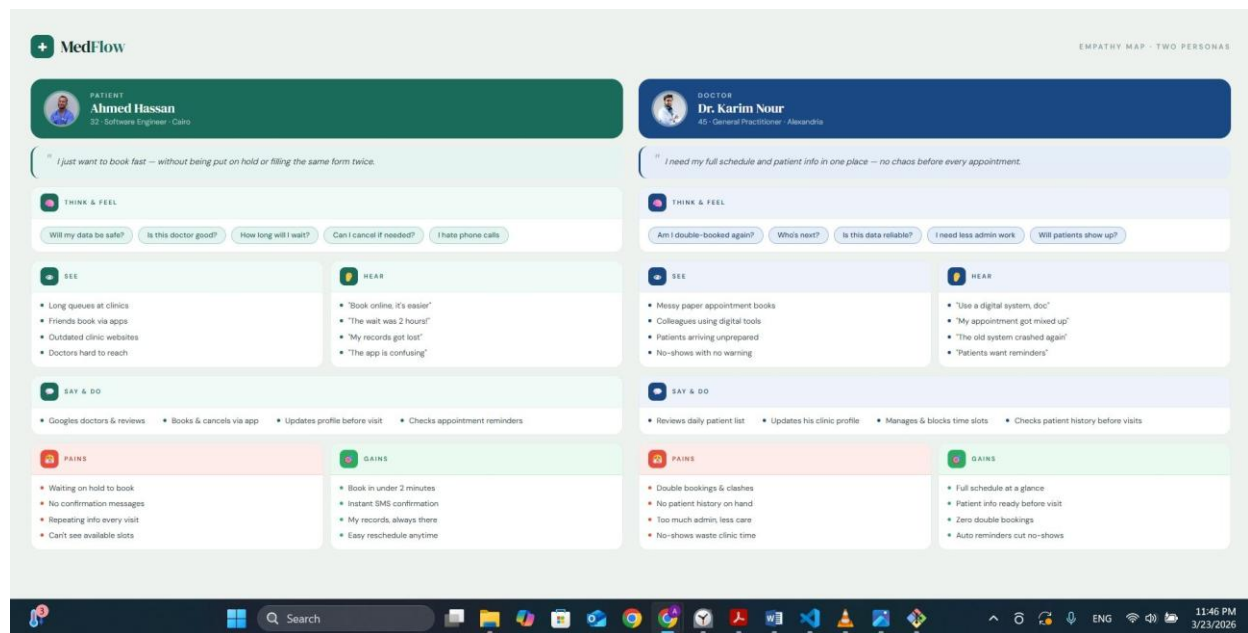


Figure — Empathy Map: Patient (left) & Doctor (right)

User Personas

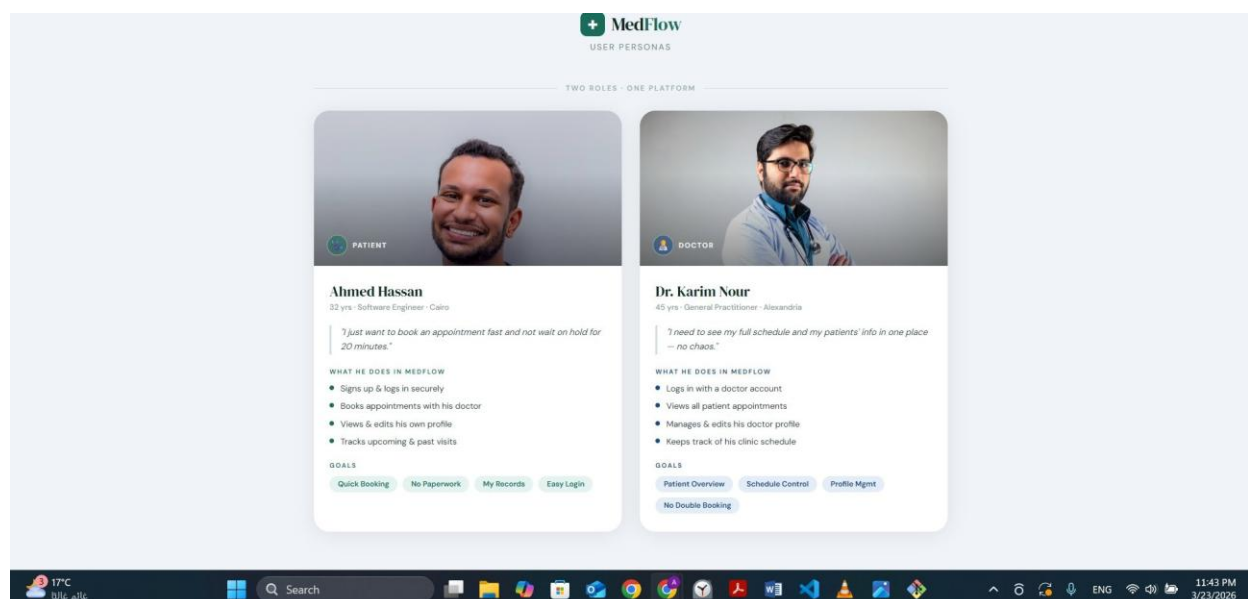


Figure — User Personas: Ahmed Hassan (Patient) & Dr. Karim Nour (Doctor)

Detailed personas were created to keep design decisions grounded in real user needs.

Attribute	Ahmed Hassan — Patient	Dr. Karim Nour — Doctor
Age / Role	32 · Software Engineer · Cairo	45 · General Practitioner · Alexandria
Quote	"Book fast, no hold time."	"Full schedule in one place — no chaos."
Goals	Quick booking, no paperwork, easy login	Patient overview, schedule control, no double-booking
Pains	Waiting on hold, no confirmation, repeating info	Double bookings, no patient history, admin overload
Gains	Book in 2 min, instant SMS, easy reschedule	Full schedule glance, patient info ready, auto reminders

Stage II — Define

Problem Statements

Four pain-point categories were identified from research and mapped to concrete problem statements.

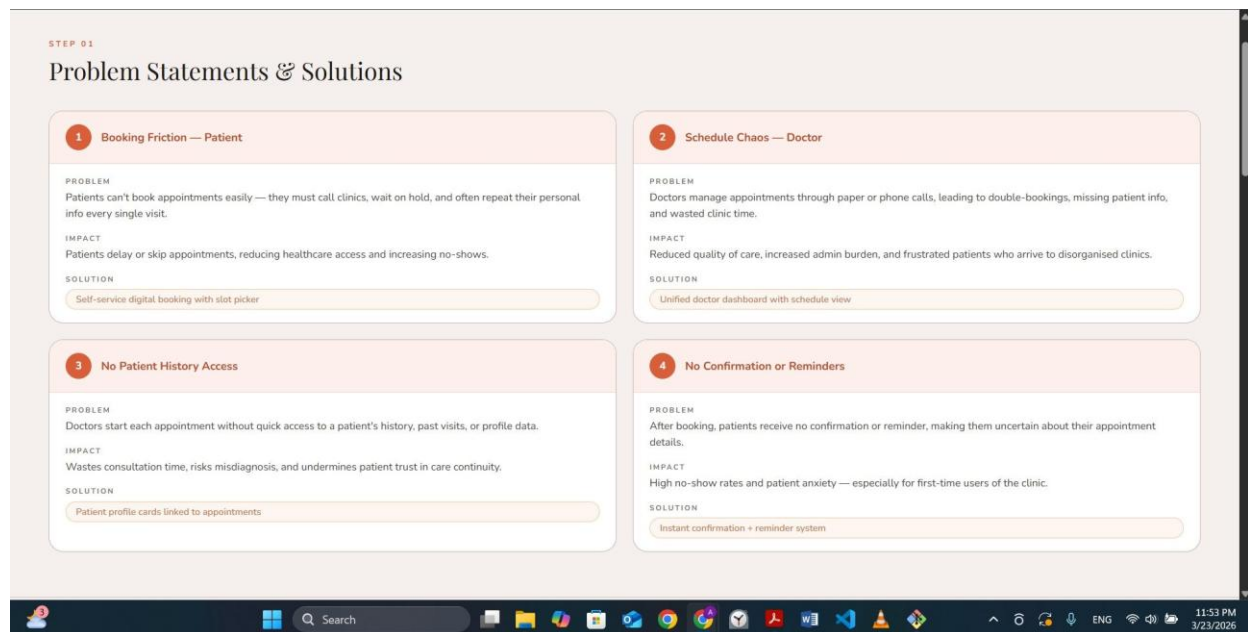


Figure — Problem Statements & Solutions

1 Booking Friction — Patient

PROBLEM

Patients must call clinics, wait on hold, and repeat personal info every visit.

IMPACT

Patients delay or skip care, increasing no-shows.

SOLUTION

Self-service digital booking with slot picker

2 Schedule Chaos — Doctor

PROBLEM

Doctors manage appointments via paper or phone, causing double-bookings and missing patient info.

IMPACT

Reduced care quality and high admin burden.

SOLUTION

Unified doctor dashboard with schedule view

3 No Patient History Access

PROBLEM

Doctors start each appointment without quick access to patient history or profile data.

IMPACT

Wasted consultation time and risk of misdiagnosis.

SOLUTION

Patient profile cards linked to appointments

4 No Confirmation or Reminders

PROBLEM

After booking, patients receive no confirmation or reminder.

IMPACT

High no-show rates and patient anxiety.

SOLUTION

Instant confirmation + reminder notification system

Stage III — Ideate

Goal Statements

- G1 Reduce patient booking time to under 2 minutes by providing a self-service digital slot picker with instant confirmation.
- G2 Give doctors a unified dashboard so they can view all appointments, patient info, and schedule without double-bookings.
- G3 Ensure patients receive instant booking confirmation and pre-appointment reminders to reduce no-show rates.
- G4 Allow patients to maintain a persistent profile so they never repeat personal information across visits.

Selected Solution & Feature List

Selected Solution: A responsive Angular web app (MedFlow) with a Spring Boot REST backend and PostgreSQL database, providing separate patient and doctor portals within a single unified platform.

Why this approach? Existing solutions (Zocdoc, Vezeeta) are either region-locked, expensive, or lack a real-time doctor dashboard. MedFlow is open, self-hosted, and combines both patient booking and doctor management in one codebase with a consistent design system.

Feature	Description
Authentication	Sign In / Sign Up with Patient or Doctor role selection, Google OAuth
Patient Portal	Dashboard, upcoming & past appointments, quick actions, online doctors list
Doctor Portal	Doctor dashboard with today's patients, stats (patients, reviews, rating, experience)
Appointment Booking	Calendar-based slot picker, confirmation, cancel action
My Appointments	Tabbed view — Today / Upcoming / History for both roles
Notifications	Bell icon with badge count, slide-in notification panel, clear all
Doctor Discovery	Filter by specialty & search, doctor profile with working hours & fee
Patient List	Doctor-side patient registry with search, total count, patient profiles

Profile Management	Edit profile for both roles, medical baseline (age, blood, gender) for patients
Navigation	Responsive sidebar (desktop) + bottom navbar (mobile) for both roles

Stage IV — Wireframes & Site Map

High-fidelity wireframes were produced for all 12 screens before development. The site map below reflects the navigation structure implemented in the final app.

Screens covered: Login, Register, Home, Profile, Find Doctor, Doctor Profile, Book Appointment, My Appointments, Doctor Dashboard, Doctor Appointments, Doctor Profile (edit), Patients List.

(Wireframe link: see project GitHub repository)

SECTION 02

Frontend

MedFlow is built with Angular 17 (standalone components), TypeScript, and a custom design system inspired by warm, earthy healthcare tones. The app is fully responsive, adapting between a mobile bottom-navbar layout and a desktop sidebar layout.

Technologies Used

Layer	Technology	Purpose
Frontend Framework	Angular 17 (Standalone)	Component-based SPA
Language	TypeScript 5	Type-safe frontend code
Styling	CSS3 + CSS Variables	Design tokens, responsive layout
Icons	Font Awesome 6	UI icons throughout the app
HTTP	Angular HttpClient	REST API communication
Routing	Angular Router	Client-side navigation & guards
State / Auth	SessionStorage + AuthService	User session management
Backend	Spring Boot (Java)	REST API server
Database	PostgreSQL	Persistent data storage
Build Tool	Angular CLI / Vite	Dev server & production builds

Design Colors

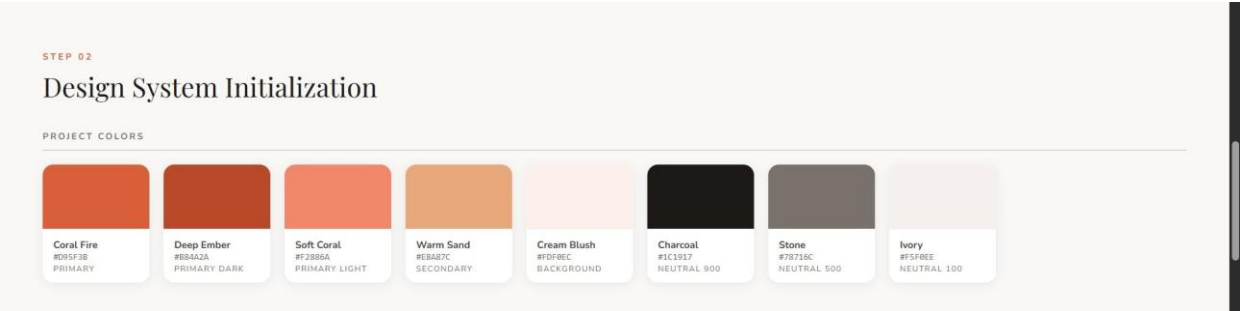


Figure — MedFlow Design System Colors

Token	Name	Hex	Role
--primary	Coral Fire	#D95F3B	Primary action, CTAs, headers
--primary-dark	Deep Ember	#B84A2A	Hover states, active elements
--primary-light	Soft Coral	#F2886A	Highlights, badges
--secondary	Warm Sand	#E8A87C	Secondary accents
--bg	Cream Blush	#FDF0EC	Page background
--neutral-900	Charcoal	#1C1917	Primary text
--neutral-500	Stone	#78716C	Secondary text, labels
--neutral-100	Ivory	#F5F0EE	Card backgrounds, table rows

Typography

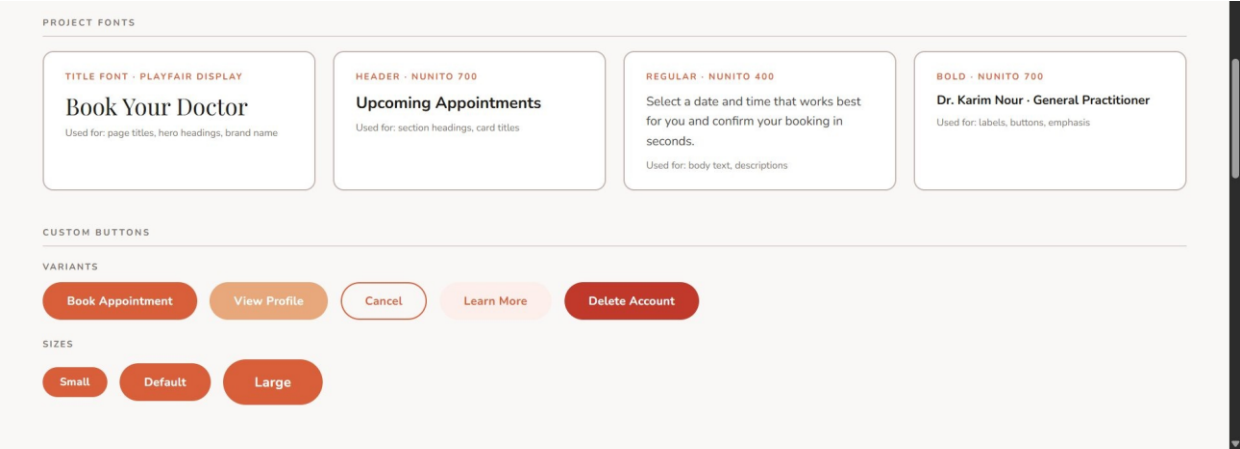


Figure — Project Fonts & Custom Buttons

Role	Font	Size	Weight	Usage
Display / Logo	Playfair Display	28–36px	Bold 700	App name, hero headings
Section Titles	Nunito 700	20–26px	SemiBold 600	Page titles, card headers
Body Text	Nunito 400	13–15px	Regular 400	Paragraphs, descriptions
Labels & Badges	Nunito 700	10–12px	Bold 700	Status chips, meta info
Form Inputs	Nunito	14px	Regular 400	Input fields, placeholders

Custom Components

The following reusable Angular standalone components are shared across the app. Each is encapsulated with its own HTML template and scoped CSS.

Component	Style / Description	Used In
Sidebar	Fixed left panel on desktop ($\geq 1024\text{px}$); hidden on mobile. Shows logo, nav links with active highlight, and user role. Width: 240px, background white, primary-colored active states.	Home, Doctor Dashboard, Appointments, Doctors, Patients, Profile (all desktop views)
Navbar	Sticky bottom bar on mobile ($< 1024\text{px}$); hidden on desktop. Contains 4–5 icon+label tab items. Active tab highlighted in primary color. Accepts [isDoctor] and [activePage] inputs.	All pages — patient and doctor portals
Notification Panel	Full-height slide-in panel from the right. Semi-transparent overlay. Displays notification cards (unread highlighted in primary tint). Emits (close) event. Bell icon shows red badge.	Home, Doctor Dashboard
Alert / Toast	Inline alert with icon, message, and optional dismiss button. Variants: success (green), error (red/coral), info (blue). Auto-dismisses after 4s or on user click.	Booking confirmation, form validation, profile update success

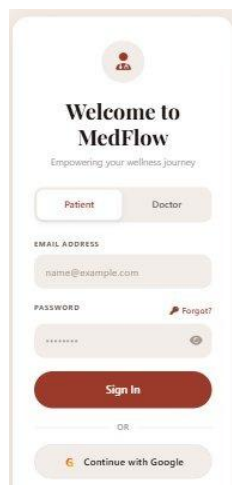
Custom Buttons

Name	CSS Class	Style	Used In
Primary Button	.btn-primary	Background: Coral Fire #D95F3B; text white; border-radius 999px; hover darkens to Deep Ember.	Sign In, Book Appointment, Confirm, Edit Profile
Outline Button	.btn-outline	Transparent background; 1px border in primary color; text primary; hover fills with primary.	Cancel, secondary actions in forms
Icon Button	.icon-btn	Circular 42px; white background; primary-colored icon; box-shadow; badge position.	Notification bell, profile nav, back arrow
Quick Action Btn	.quick-action-btn	Square card shape; bg-muted fill; column flex (icon + label); hover lifts with translateY(-2px).	Home quick actions grid
Tab Button	.tab-btn	Pill-shaped; inactive = muted bg; active = primary bg white text; groups of 2–3.	Login role selector, Appointments tabs
Status Badge	.appointment-status	Pill shape; UPCOMING=blue tint, COMPLETED=green tint, CANCELLED=red tint.	Appointment cards throughout the app

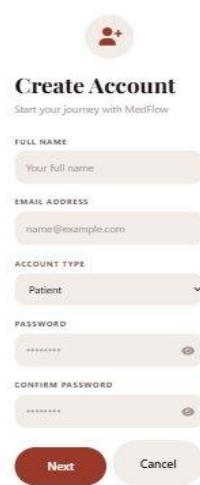
UI Design Screenshots

All screens below are captured from the live MedFlow application, showing the mobile-first responsive layout.

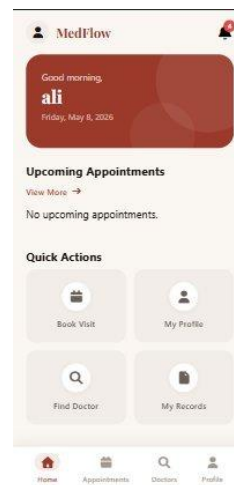
Patient Screens



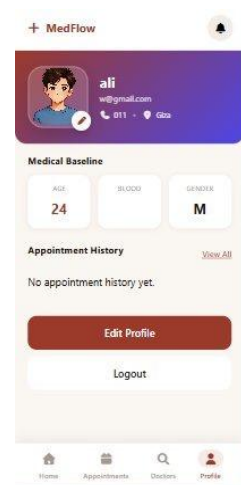
Login



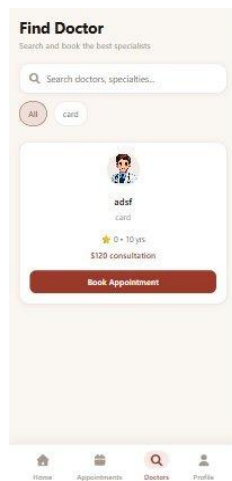
Register



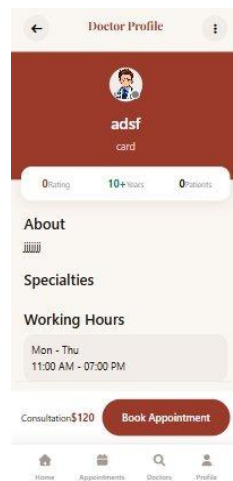
Home



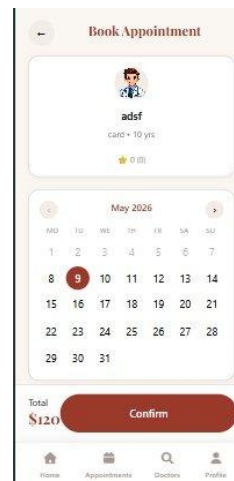
Patient Profile



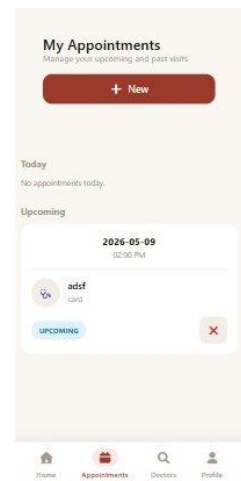
Find Doctor



Doctor Profile

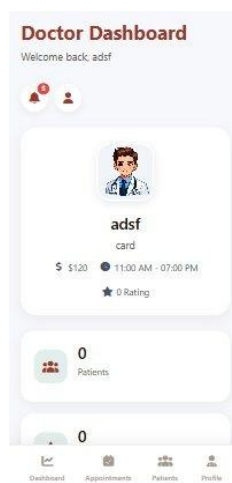


Book Appointment

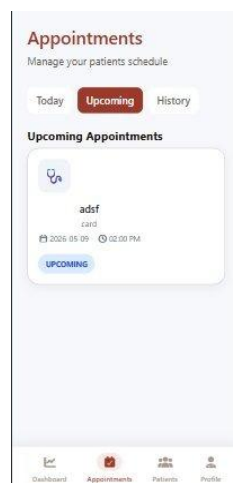


My Appointments

Doctor Screens



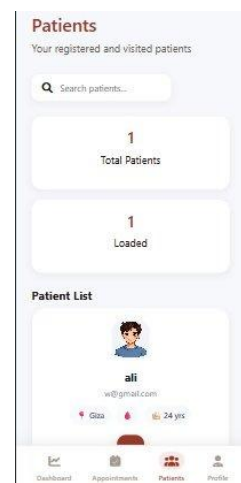
Doctor Dashboard



Doctor Appointments



Doctor Profile (edit)



Patients List

Responsiveness

MedFlow uses CSS media queries and flexible layouts to adapt between mobile (< 768px), tablet (768–1023px), and desktop (≥ 1024px) viewports.

Element	Behaviour
Bottom Navbar ↔ Sidebar	On mobile, renders a sticky bottom tab bar. On desktop (≥1024px), it is hidden and replaced by a fixed sidebar on the left.
Hero Banner	Padding and font sizes increase on desktop. The home content-wrap switches to a 2-column CSS Grid layout.
Stats Grid	Doctor dashboard stats display as 4 columns on desktop, 2 on tablet, 1 on mobile.
Doctor Cards Grid	"Top Specialists" section uses 1 column on mobile and 3 columns on desktop.
Appointment Cards	Flex direction switches from row (desktop) to column (mobile) so status badges align correctly.
Forms	All form inputs are full-width; the role toggle tabs shrink gracefully on small screens.

SECTION 03

Backend

MedFlow's backend is built with Spring Boot (Java) and a PostgreSQL relational database, exposed as a RESTful API consumed by the Angular frontend via Angular HttpClient.

Database Records

Representative snippets of database records from different parts of the application, illustrating how data is persisted across the key entities.

Users / Patients Table

Stores patient accounts including personal details and medical baseline (age, blood type, gender, location). The role field differentiates between PATIENT and DOCTOR accounts.

Doctors Table

Extends user records with doctor-specific fields: specialty, about text, working hours, consultation fee, years of experience, and aggregate stats (rating, patient count, review count).

Appointments Table

Records each booking with foreign keys to both patient and doctor, appointment date/time, and a status enum (UPCOMING, COMPLETED, CANCELLED). This table drives all appointment views for both portals.

Notifications Table

Stores per-user notification entries with a read flag, message body, and timestamp. The notification panel queries this table on load and marks entries as read on dismiss.

Dynamic Content

All displayed data in MedFlow is fetched dynamically from the PostgreSQL database via REST API calls. Key dynamic content flows:

- Patient Home: upcoming appointments and quick stats are fetched on login and refreshed on navigation.
- Doctor Discovery: doctor cards are retrieved from /api/doctors with optional specialty filter and search query parameters.
- Appointment Booking: available time slots are fetched per doctor and date, preventing double-booking at the database constraint level.
- Notifications: unread notification count drives the badge on the bell icon; the panel list is fetched and marked read via PATCH requests.
- Doctor Dashboard: today's patient count, total appointments, rating, and experience are computed server-side and returned as a stats DTO.
- Patient List (Doctor view): paginated patient registry is fetched with a search parameter for real-time filtering.

Database Screenshots

The following screenshots are taken directly from pgAdmin 4 connected to the live PostgreSQL 18 instance (medical_db). They confirm the structure and sample data stored across the five core tables that power the MedFlow application.

Figure 1 — Patients Table

The screenshot shows the pgAdmin 4 interface with the 'medical_db/postgres@PostgreSQL 18*' connection selected. The 'Query' tab is active, displaying a SQL query: `select * from patients`. The 'Data Output' tab shows the results of the query, displaying 12 rows of patient data. The table structure is as follows:

id	timestamp without time zone (6)	blood	gender	avatar	phone	name	email	location	password
		character varying (10)	character varying (10)	character varying (50)	character varying (50)	character varying (150)	character varying (255)	character varying (255)	character varying (255)
1	5-05-09 01:51:27.725594	1 O+	male	./././assets/avatars/boy...	01012345678	Ahmed Ali	ahmed.ali@medflow.com	Giza	\$2a\$10\$wz0hB9Uc1CF4j0
2	5-05-09 01:51:27.918503	2 A+	female	./././assets/avatars/girl.p...	01123456789	Sara Adel	sara.adel@medflow.com	Cairo	\$2a\$10\$1w26bWzmXwS0
3	5-05-09 01:51:28.03373	3 B+	male	./././assets/avatars/boy...	01234567890	Omar Hassan	omar.hassan@medflow.com	Alexandria	\$2a\$10\$wy16C0EIMcszBv
4	5-05-09 01:51:28.151398	4 AB+	female	./././assets/avatars/girl.p...	01098765432	Mona Farid	mona.farid@medflow.com	Mansoura	\$2a\$10\$bousU93rxV4tyX
5	5-05-09 01:51:28.265698	5 O-	male	./././assets/avatars/boy...	01155667788	Youssef Nabil	youssef.nabil@medflow.com	Tanta	\$2a\$10\$71Jr9611zqPCaX
6	5-05-09 01:51:28.383359	6 A-	female	./././assets/avatars/girl.p...	01266778899	Hana Mahmoud	hana.mahmoud@medflow...	Zagazig	\$2a\$10\$UanzQdd0NkV0b
7	5-05-09 01:51:28.50053	7 B-	male	./././assets/avatars/boy...	01024681357	Khaled Samir	khaled.samir@medflow.com	Beni Suef	\$2a\$10\$Q03SHNVjmdEM
8	5-05-09 01:51:28.618077	8 O+	female	./././assets/avatars/girl.p...	01122334455	Dina Youssef	dina.youssef@medflow.com	Fayoum	\$2a\$10\$Bc7QQWkuIVwK0
9	5-05-09 01:51:28.733617	9 A+	male	./././assets/avatars/boy...	01233445566	Mostafa Adel	mostafa.adel@medflow.com	Ismailia	\$2a\$10\$zUhfQqXHC9Qp
10	5-05-09 01:51:28.849562	10 AB-	female	./././assets/avatars/girl.p...	01055667799	Nourhan Tarek	nourhan.tarek@medflow.co...	Port Said	\$2a\$10\$ExTUJ8nRTQYd7C
11	5-05-09 01:51:28.967656	11 O+	male	./././assets/avatars/boy...	01177889900	Mahmoud Gamal	mahmoud.gamal@medflow...	Helwan	\$2a\$10\$GKZQh52fD.W1R
12	5-05-09 01:51:29.084424	12 B+	female	./././assets/avatars/girl.p...	01299001122	Salma Ibrahim	salma.ibrahim@medflow.c...	6th of October	\$2a\$10\$EoLAbf7PWPWg

Total rows: 12 Query complete 00:00:00.154 CRLF Ln 2, Col 1

The patients table stores 12 patient accounts with personal details (name, email, phone, location, blood type, gender) and bcrypt-hashed passwords. Avatar paths reference local asset files.

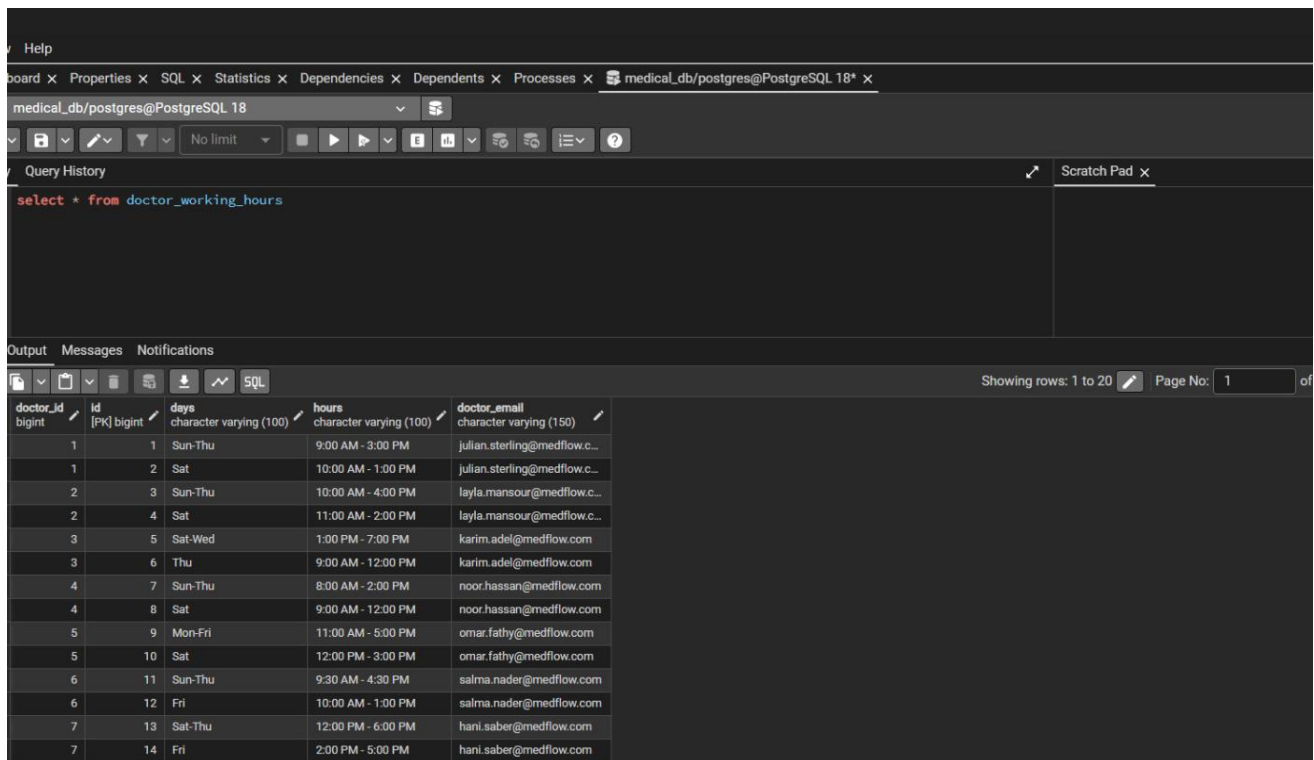
Figure 2 — Appointments Table

The screenshot shows a PostgreSQL database interface with the 'appointments' table selected. The table has 10 columns and 10 rows of data. The columns are: consultation_fee (numeric), experience (integer), is_online (boolean), patients_count (integer), rating (numeric), reviews (integer), created_at (timestamp), id (bigint), specialty (character varying), email (character varying), and name (character varying). The data shows 10 rows of appointment records for various doctors.

consultation_fee	experience	is_online	patients_count	rating	reviews	created_at	id	specialty	email	name
120.00	12	true	2400	4.9	120	2026-05-09 01:51:29.206517	1	Cardiologist	julian.sterling@medflow.c...	Dr. Julian Sterling
110.00	9	true	1820	4.8	96	2026-05-09 01:51:29.373859	2	Dermatologist	layla.mansour@medflow.c...	Dr. Layla Mansour
150.00	15	false	3180	4.7	143	2026-05-09 01:51:29.501353	3	Orthopedic Surgeon	karim.adel@medflow.com	Dr. Karim Adel
95.00	8	true	1490	4.9	88	2026-05-09 01:51:29.622876	4	Pediatrician	noor.hassan@medflow.com	Dr. Noor Hassan
140.00	13	false	2110	4.6	104	2026-05-09 01:51:29.747915	5	Neurologist	omar.fathy@medflow.com	Dr. Omar Fathy
130.00	10	true	2265	4.8	111	2026-05-09 01:51:29.871763	6	Gynecologist	salma.nader@medflow.com	Dr. Salma Nader
100.00	11	true	1735	4.5	79	2026-05-09 01:51:29.995178	7	ENT Specialist	hani.saber@medflow.com	Dr. Hani Saber
145.00	14	false	2750	4.9	132	2026-05-09 01:51:30.119657	8	Endocrinologist	dina.farouk@medflow.com	Dr. Dina Farouk
115.00	7	true	1215	4.7	67	2026-05-09 01:51:30.240031	9	Psychiatrist	mostafa.rami@medflow.c...	Dr. Mostafa Rami
90.00	16	true	3425	4.8	155	2026-05-09 01:51:30.364401	10	Family Medicine	hala.youssef@medflow.co...	Dr. Hala Youssef

The appointments table holds 9 booking records linking patients and doctors via foreign keys. Each row captures the appointment date, time slot (e.g., 10:00 AM), duration in minutes, specialty, doctor name, and a status enum (UPCOMING / COMPLETED / CANCELLED). Icon and icon_bg columns supply the card visual in the frontend.

Figure 3 — Doctors Table



The screenshot shows a PostgreSQL query editor interface. The top menu bar includes Help, board, Properties, SQL, Statistics, Dependencies, Dependents, Processes, and a connection tab for 'medical_db/postgres@PostgreSQL 18'. Below the menu is a toolbar with various icons for query execution and management. The main area is divided into 'Query History' and 'Scratch Pad'. The 'Query History' tab shows a SQL query: `select * from doctor_working_hours`. Below this is the 'Output' tab, which displays the results of the query in a table format. The table has 5 columns: `doctor_id`, `id`, `days`, `hours`, and `doctor_email`. The results show 14 rows of data, representing 10 doctor profiles with their working hours and email addresses.

doctor_id	id	days	hours	doctor_email
1	1	Sun-Thu	9:00 AM - 3:00 PM	julian.sterling@medflow.c...
1	2	Sat	10:00 AM - 1:00 PM	julian.sterling@medflow.c...
2	3	Sun-Thu	10:00 AM - 4:00 PM	layla.mansour@medflow.c...
2	4	Sat	11:00 AM - 2:00 PM	layla.mansour@medflow.c...
3	5	Sat-Wed	1:00 PM - 7:00 PM	karim.adel@medflow.com
3	6	Thu	9:00 AM - 12:00 PM	karim.adel@medflow.com
4	7	Sun-Thu	8:00 AM - 2:00 PM	noor.hassan@medflow.com
4	8	Sat	9:00 AM - 12:00 PM	noor.hassan@medflow.com
5	9	Mon-Fri	11:00 AM - 5:00 PM	omar.fathy@medflow.com
5	10	Sat	12:00 PM - 3:00 PM	omar.fathy@medflow.com
6	11	Sun-Thu	9:30 AM - 4:30 PM	salma.nader@medflow.com
6	12	Fri	10:00 AM - 1:00 PM	salma.nader@medflow.com
7	13	Sat-Thu	12:00 PM - 6:00 PM	hani.saber@medflow.com
7	14	Fri	2:00 PM - 5:00 PM	hani.saber@medflow.com

The doctors table stores 10 doctor profiles. Aggregate fields (rating, reviews, patients_count, experience) are updated server-side. The consultation_fee (numeric) and is_online flag drive the doctor discovery cards. Each record links to an avatar asset path and a bio text column (truncated in view).

Figure 4 — Notifications Table

read	appointment_id	created_at	id	read_at	recipient_id	recipient_role	icon	title
boolean	bigint	timestamp without time zone (6)	[PK] bigint	timestamp without time zone (6)	bigint	character varying (20)	character varying (50)	character varying (200)
false	1	2026-05-09 03:16:46.166525	1	[null]	1	PATIENT	fa-calendar-check	Appointment booked with Dr. Dr. Julian Sterling on 2026-05-09 at 10:00...
false	1	2026-05-09 03:16:46.173577	2	[null]	1	DOCTOR	fa-calendar-plus	New appointment booked by Ahmed Ali on 2026-05-09 at 10:00 AM
false	2	2026-05-09 03:16:46.193663	3	[null]	2	PATIENT	fa-calendar-check	Appointment booked with Dr. Dr. Layla Mansour on 2026-05-09 at 11:30...
false	2	2026-05-09 03:16:46.194661	4	[null]	2	DOCTOR	fa-calendar-plus	New appointment booked by Sara Adel on 2026-05-09 at 11:30 AM
false	3	2026-05-09 03:16:46.213309	5	[null]	3	PATIENT	fa-calendar-check	Appointment booked with Dr. Dr. Hani Saber on 2026-05-09 at 1:00 PM
false	3	2026-05-09 03:16:46.214821	6	[null]	3	DOCTOR	fa-calendar-plus	New appointment booked by Omar Hassan on 2026-05-09 at 1:00 PM
false	3	2026-05-09 03:16:46.235634	7	[null]	3	PATIENT	fa-circle-check	Appointment completed with Dr. Dr. Hani Saber
false	3	2026-05-09 03:16:46.238697	8	[null]	7	DOCTOR	fa-circle-check	Appointment completed for Omar Hassan
false	4	2026-05-09 03:16:46.26898	9	[null]	4	PATIENT	fa-calendar-check	Appointment booked with Dr. Dr. Salma Nader on 2026-05-10 at 9:00 AM
false	4	2026-05-09 03:16:46.269967	10	[null]	6	DOCTOR	fa-calendar-plus	New appointment booked by Mona Farid on 2026-05-10 at 9:00 AM
false	5	2026-05-09 03:16:46.286592	11	[null]	5	PATIENT	fa-calendar-check	Appointment booked with Dr. Dr. Dina Farouk on 2026-05-10 at 2:00 PM
false	5	2026-05-09 03:16:46.287734	12	[null]	8	DOCTOR	fa-calendar-plus	New appointment booked by Youssef Nabil on 2026-05-10 at 2:00 PM
false	5	2026-05-09 03:16:46.294735	13	[null]	5	PATIENT	fa-calendar-xmark	Appointment cancelled with Dr. Dr. Dina Farouk
false	5	2026-05-09 03:16:46.297774	14	[null]	8	DOCTOR	fa-calendar-xmark	Appointment cancelled by Youssef Nabil
false	6	2026-05-09 03:16:46.314995	15	[null]	13	PATIENT	fa-calendar-check	Appointment booked with Dr. Dr. Noor Hassan on 2026-05-11 at 12:30 ...
false	6	2026-05-09 03:16:46.316555	16	[null]	4	DOCTOR	fa-calendar-plus	New appointment booked by Laila Hamed on 2026-05-11 at 12:30 PM
false	7	2026-05-09 03:16:46.334544	17	[null]	12	PATIENT	fa-calendar-check	Appointment booked with Dr. Dr. Mostafa Bami on 2026-05-12 at 4:00 P...

The notifications table contains 22 entries (2 per appointment — one for the patient, one for the doctor). The read boolean defaults to false and is flipped to true when the user dismisses the panel. The recipient_role column (PATIENT / DOCTOR) allows the API to scope queries per user type. The title column holds the human-readable notification message shown in the slide-in panel.

Figure 5 — Doctor Working Hours Table

The screenshot shows a PostgreSQL database interface with the 'appointments' table selected. The query history shows 'select * from appointments'. The output pane displays the following data:

appointment_date	duration	created_at	doctor_id	id	patient_id	appointment_time	icon	icon_bg	status
2026-05-09	30	2026-05-09 03:16:46.158929	1	1	1	10:00 AM	fa-stethoscope	#e8f3ff	UPCOMING
2026-05-09	45	2026-05-09 03:16:46.19166	2	2	2	11:30 AM	fa-face-smile	#f3e8ff	UPCOMING
2026-05-09	30	2026-05-09 03:16:46.21131	7	3	3	1:00 PM	fa-ear-listen	#eef2ff	COMPLETED
2026-05-10	40	2026-05-09 03:16:46.266861	6	4	4	9:00 AM	fa-person-pregnant	#fff1f2	UPCOMING
2026-05-10	30	2026-05-09 03:16:46.28377	8	5	5	2:00 PM	fa-vial-circle-check	#ecfeff	CANCELLED
2026-05-11	30	2026-05-09 03:16:46.313485	4	6	13	12:30 PM	fa-baby	#f0fdf4	UPCOMING
2026-05-12	50	2026-05-09 03:16:46.332038	9	7	12	4:00 PM	fa-brain	#fff7ed	UPCOMING
2026-05-08	30	2026-05-09 03:16:46.350647	3	8	7	3:30 PM	fa-bone	#f8faff	UPCOMING
2026-05-13	30	2026-05-09 03:18:41.326415	3	9	1	02:00 PM	[null]	[null]	UPCOMING

The `doctor_working_hours` table holds 20 rows (2 schedule slots per doctor). Each row stores the working days (e.g., Sun–Thu, Sat) and the corresponding time range (e.g., 9:00 AM – 3:00 PM). This data is consumed by the doctor profile page and the appointment booking slot picker to display availability and prevent out-of-hours bookings.

Survey Methodology

Survey Overview		Behavior
Survey Period	April 15 – May 5, 2026	
Total Respondents	250 (150 Patients + 100 Doctors)	
Locations	Cairo, Alexandria, Giza, Mansoura	
Method	Online questionnaire + In-clinic interviews	
Objective	Identify scheduling pain points and opportunities for digital appointment booking platform	

Key Statistics

250	150	100	94%
Total Respondents	Patients Surveyed	Doctors Surveyed	Response Rate

Patient Survey Results (n=150)

Q1:How do you currently book doctor appointments?

Phone calls	42%	63
Walk-in visits	35%	52
WhatsApp / Messaging	18%	27
Online platform	5%	8

Q2:How long does it typically take to book an appointment?

Less than 5 minutes	8%	12
5-15 minutes	12%	18
15-30 minutes	28%	42
More than 30 minutes / Multiple calls	52%	78

Q3:What frustrates you most about booking appointments?

Long wait times on phone	48%	72
No confirmation / unclear appointment details	38%	57
Limited availability / long wait for slots	32%	48
Can't see doctor credentials / background	22%	33

Q4:Would you use an online booking platform if it offered these features?

Yes - Very likely	72%	108
Yes - Somewhat likely	20%	30
Neutral	6%	9
Unlikely	2%	3

Q5:What features are most important in a booking app? (Top 3)

Instant confirmation & reminders	85%	128
View doctor info (specialty, experience, ratings)	78%	117
See available time slots in real-time	82%	123
Easy rescheduling / cancellation	65%	97

Doctor Survey Results (n=100)

Q1:How do you currently manage appointment schedules?

Paper-based / handwritten notes	38%	38
Phone calls only	42%	42
Basic Excel/spreadsheet	15%	15
Dedicated clinic management software	5%	5

Q2:What's your main challenge with appointment management?

Double bookings / scheduling conflicts	52%	52
No-show patients wasting time	45%	45
Staff time spent on phone booking	35%	35
Difficulty retrieving patient history	28%	28

Q3:How often do patients no-show for appointments?

Rarely (less than 5%)	15%	15
Sometimes (5-15%)	22%	22
Often (15-30%)	35%	35
Frequently (30%+)	28%	28

Q4:Would you adopt a digital booking platform?

Definitely - need it urgently	68%	68
Probably - if easy to use	24%	24
Maybe - concerned about complexity	6%	6
No - prefer current method	2%	2

Q5: Most valuable platform features for doctors? (Top 3)

Automated schedule blocking & no-show handling	81%	81
Patient history & medical records access	76%	76
Automatic SMS/email reminders to patients	72%	72
Reduced admin staff burden	65%	65

Pain Points & Gains Analysis

Top Patient Pain Points

- 52% spend 30+ min booking (high friction)
- 48% frustrated by long phone waits
- 38% don't get clear confirmation
- 32% struggle finding available slots
- 22% can't verify doctor credentials

Expected Patient Gains

- 72% very likely to use online booking
- 85% want instant confirmation & reminders
- 82% need real-time slot visibility
- 78% value doctor info (specialty, ratings)
- Projected: 60-70% booking time reduction

Top Doctor Pain Points

- 52% report double-booking conflicts
- 45% suffer from no-show patients (15-30%+ rate)
- 80% use manual/paper-based scheduling
- 35% lose staff time on phone bookings
- 28% difficulty accessing patient history

Expected Doctor Gains

- 68% urgently want digital solution
- 81% value automated scheduling & no-show handling
- 76% want patient history integration
- 72% need SMS/email reminder system
- Projected: Reduce no-shows by 40-50%